Day 8 Task — Capture Analysis

Time needed	Deadline	Where to submit
45 minutes	Before the Day 9 session	Portal → Day 8 → Upload

Download **day8_capture.pcap** from the portal (attached to this task) and open it in **Wireshark**. It is a short capture from a small office network, **10.0.0.0/24**. Three things in it are suspicious — your job is to find them using filters, Follow Stream and the Statistics menus. Take screenshots of your evidence as you go.

This is a safe, lab-generated capture. Analysing a provided .pcap is always legal.

Warm-up — get your bearings

Question	Your answer
How many packets are in the capture? (bottom bar)	
Statistics → Protocol Hierarchy: which protocols appear?	
Statistics → Endpoints: how many IPv4 addresses are involved?	

Finding 1 — The plaintext login

Someone logged in over HTTP, so the credentials are in clear text. Filter, then right-click the POST packet → Follow → TCP Stream.

```
Display filter: http.request.method == "POST"
```

Question	Your answer
Which client IP sent the login?	
Which server IP received it?	
What is the username?	
What is the password? (it's URL-encoded — %40 = @)	
Why was this readable, and what one change would hide it?	

Finding 2 — The port scan

One host is scanning the server. SYN with no ACK = a probe; the server replies SYN-ACK for open ports, RST for closed.

```
Display filter: tcp.flags.syn == 1 && tcp.flags.ack == 0
```

Question	Your answer
Which IP is doing the scanning (the attacker)?	

Question	Your answer
Which IP is being scanned?	
Roughly how many ports were probed?	
Which ports replied SYN-ACK (open)? (change filter to tcp.flags.syn==1 && tcp.flags.ack==1)	
What are those open ports normally used for?	

Finding 3 — The malware beacon (C2)

One internal host keeps contacting the same external IP at a steady rhythm. Use Statistics → Conversations → TCP tab, and sort by packets.

Question	Your answer
Which internal host is beaconing?	
Which external IP is it contacting (the C2 server)?	
On which destination port?	
Roughly how often does it beacon (seconds between connections)?	
How is this different from the normal HTTPS traffic in the capture?	

Step 4 — Write the mini incident report (4–6 lines)

Summarise for a non-technical manager: what you found (the three issues), which machine is compromised, and the three things you'd do first. Use the Who/What/Fix style from this week.

How to submit

- One file: Google Doc or PDF with your answers and screenshots of the key evidence
- Must contain: the four tables, screenshots for each finding, and the incident report
- Upload under Day 8 on the portal before the Day 9 session

Marks (out of 10)

Part	Marks
Finding 1 — login found with username and password	3
Finding 2 — scanner identified and open ports listed	3
Finding 3 — beaconing host and C2 IP identified	3
Step 4 — clear incident summary	1

Counts toward your lab journal (30%). Solutions revealed at the start of Day 9. Tomorrow we take the file that a scan like this would drop and analyse it on VirtualTotal.